

EV Power - Lab 4 Project Report

Example Solution 1

Part 0: libraries

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.1
✓ ggplot2    4.0.0    ✓ tibble     3.2.1
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.0.4
— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(readr)
library(stringr)
library(scales)
```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

```
library(maps)
```

Attaching package: 'maps'

The following object is masked from 'package:purrr':

map

```
library(leaflet)
library(viridis)
```

Loading required package: viridisLite

Attaching package: 'viridis'

The following object is masked from 'package:maps':

unemp

The following object is masked from 'package:scales':

viridis_pal

Part 1: Defining Research Question

Chosen Question:

The main research question of this project is:

Do electric vehicles actually use clean electricity sources?

To explore this, I propose the following sub-questions:

1. How has the share of renewable energy changed from 2021 to 2023 across U.S. states?
2. Are states with more EV registrations powered by cleaner energy sources?
3. Do states with higher renewable energy shares have lower or higher average electricity prices?

Part 2: Data Preparation and Cleaning

```
getwd()
```

```
[1] "C:/ev-power-qiguofeng-spec-main"
```

```
# Renewable energy use by year

abbr_to_name <- setNames(state.name, state.abb)
abbr_to_name <- c(abbr_to_name, c(DC = "District Of Columbia"))
to_name <- function(s){
  a <- toupper(str_trim(s))
  out <- ifelse(a %in% names(abbr_to_name), abbr_to_name[a], str_to_title(a))
  str_squish(out)
}
```

```
# ----- READ -----
renew_2021 <- read_csv("C:/ev-power-qiguofeng-spec-main/ev-power-qiguofeng-spec-main/data/renew
renew_2022 <- read_csv("C:/ev-power-qiguofeng-spec-main/ev-power-qiguofeng-spec-main/data/renew
renew_2023 <- read_csv("C:/ev-power-qiguofeng-spec-main/ev-power-qiguofeng-spec-main/data/renew

total_2021 <- read_csv("C:/ev-power-qiguofeng-spec-main/ev-power-qiguofeng-spec-main/data/tota
total_2022 <- read_csv("C:/ev-power-qiguofeng-spec-main/ev-power-qiguofeng-spec-main/data/tota
total_2023 <- read_csv("C:/ev-power-qiguofeng-spec-main/ev-power-qiguofeng-spec-main/data/tota

price_raw <- read_csv("C:/ev-power-qiguofeng-spec-main/ev-power-qiguofeng-spec-main/data/av-en
ev_raw    <- read_csv("C:/ev-power-qiguofeng-spec-main/ev-power-qiguofeng-spec-main/data/ev-re
```

New names:

- `` -> `...2`

```
# ----- RENEWABLE USE (LONG -> state-year) -----
ren_proc <- function(df, yr_col, yr_val){
  df |>
    transmute(state = to_name(state),
              renewable_use = parse_number(.data[[yr_col]]),
              year = yr_val) |>
    group_by(state, year) |>
    summarise(renewable_use = sum(renewable_use, na.rm = TRUE), .groups = "drop")
}
renew_2021 <- ren_proc(renew_2021, "renewable_use_2021", 2021)
renew_2022 <- ren_proc(renew_2022, "renewable_use_2022", 2022)
renew_2023 <- ren_proc(renew_2023, "renewable_use_2023", 2023)
renew_all  <- bind_rows(renew_2021, renew_2022, renew_2023)

# ----- TOTAL USE (WIDE -> LONG -> state-year) -----
tot_proc <- function(df, yr_val){
  st_cols <- setdiff(names(df), c("energy_source", "us"))
  df |>
    select(all_of(st_cols)) |>
    pivot_longer(everything(), names_to = "abbr", values_to = "val") |>
    group_by(abbr) |>
    summarise(total_use = sum(val, na.rm = TRUE), .groups = "drop") |>
    mutate(state = to_name(abbr), year = yr_val) |>
    select(state, year, total_use)
}
total_2021 <- tot_proc(total_2021, 2021)
total_2022 <- tot_proc(total_2022, 2022)
total_2023 <- tot_proc(total_2023, 2023)
total_all  <- bind_rows(total_2021, total_2022, total_2023)

# ----- PRICE 2023 (parse state + number from single text column) -----
price_2023 <- price_raw |>
  rename(text = 1) |>
  mutate(avg_price = parse_number(text),
         state = str_trim(str_remove(text, "[0-9].*$"))) |>
  filter(!is.na(avg_price)) |>
  mutate(state = str_to_title(state)) |>
```

```
filter(!str_detect(state, "^Total|^Us$|-")) |>
select(state, avg_price)
```

Warning: There was 1 warning in `mutate()`.

- In argument: `avg\_price = parse\_number(text)`.

Caused by warning:

- ! 1 parsing failure.

row	col	expected	actual
1	-- a number		,,,

```
# ----- EV 2023 (two columns -> state + numeric) -----
ev_2023 <- ev_raw |>
  rename(state = electric_vehicle_registrations_by_state_2023,
         ev_registrations = x2) |>
  mutate(state = str_to_title(str_trim(state)),
         ev_registrations = parse_number(ev_registrations)) |>
  filter(!is.na(ev_registrations)) |>
  select(state, ev_registrations)
```

Warning: There was 1 warning in `mutate()`.

- In argument: `ev\_registrations = parse\_number(ev\_registrations)`.

Caused by warning:

- ! 1 parsing failure.

row	col	expected	actual
2	-- a number	Count-EVs	

```
# ----- MERGE & CHECK -----
energy_2023 <- renew_all |>
  filter(year == 2023) |>
  left_join(total_all |> filter(year == 2023), by = c("state", "year")) |>
  mutate(renewable_share = renewable_use / total_use)

final_2023 <- energy_2023 |>
  left_join(ev_2023, by = "state") |>
  left_join(price_2023, by = "state")

final_2023 |> arrange(desc(renewable_share)) |> head(10)
```

A tibble: 10 × 7

	state	year	renewable_use	total_use	renewable_share	ev_registrations
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	South Dakota	2023	126540	363161	0.348	1675
2	Iowa	2023	414801	1466926	0.283	9031
3	Maine	2023	89444	328875	0.272	7377
4	Oregon	2023	236063	876891	0.269	64361
5	Washington	2023	365955	1624957	0.225	152101
6	Vermont	2023	22209	105445	0.211	7816
7	Nebraska	2023	164503	872370	0.189	6920
8	Idaho	2023	77127	421975	0.183	8501
9	California	2023	1065179	6429818	0.166	1256646
10	Minnesota	2023	223864	1601319	0.140	37050

i 1 more variable: avg_price <dbl>

Part 3: Joining / Pivoting Datasets for Analysis

```
energy_all <- renew_all |>
  left_join(total_all, by = c("state", "year")) |>
  mutate(renewable_share = renewable_use / total_use)

energy_2023 <- energy_all |>
  filter(year == 2023)

final_2023 <- energy_2023 |>
  left_join(ev_2023, by = "state") |>
  left_join(price_2023, by = "state")

final_2023 <- final_2023 |>
  mutate(
    ev_per_energy = ev_registrations / total_use,
    renewable_ratio = renewable_share * 100
  )

energy_wide <- energy_all |>
  select(state, year, renewable_share) |>
  pivot_wider(names_from = year, values_from = renewable_share, names_prefix = "share_")

write_csv(final_2023, "final_2023_cleaned.csv")

glimpse(final_2023)
```

Rows: 52

Columns: 9

```
$ state      <chr> "Alabama", "Alaska", "Arizona", "Arkansas", "Californ...
$ year       <dbl> 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023,...
$ renewable_use <dbl> 222189, 10088, 108445, 87277, 1065179, 115062, 48983,...
$ total_use   <dbl> 2265008, 746979, 1712667, 1151062, 6429818, 1359507, ...
$ renewable_share <dbl> 0.09809634, 0.01350507, 0.06331937, 0.07582302, 0.165...
$ ev_registrations <dbl> 13047, 2697, 89798, 7108, 1256646, 90083, 31557, 8435...
$ avg_price     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ ev_per_energy <dbl> 0.005760245, 0.003610543, 0.052431675, 0.006175167, 0...
$ renewable_ratio <dbl> 9.809634, 1.350507, 6.331937, 7.582302, 16.566239, 8...
```

```
summary(final_2023)
```

state	year	renewable_use	total_use
Length:52	Min. :2023	Min. : 2796	Min. : 46323
Class :character	1st Qu.:2023	1st Qu.: 65095	1st Qu.: 745604
Mode :character	Median :2023	Median : 115370	Median : 1359507
	Mean :2023	Mean : 314163	Mean : 1833083
	3rd Qu.:2023	3rd Qu.: 189718	3rd Qu.: 2134676
	Max. :2023	Max. :8187317	Max. :14059016
			NA's :1
renewable_share	ev_registrations	avg_price	ev_per_energy
Min. :0.01351	Min. : 959	Min. : NA	Min. :0.001187
1st Qu.:0.05715	1st Qu.: 8108	1st Qu.: NA	1st Qu.:0.010205

Median :0.07964	Median : 25565	Median : NA	Median :0.020146
Mean :0.10329	Mean : 69715	Mean :NaN	Mean :0.036097
3rd Qu.:0.11408	3rd Qu.: 71152	3rd Qu.: NA	3rd Qu.:0.050092
Max. :0.34844	Max. :1256646	Max. : NA	Max. :0.195440
NA's :1	NA's :1	NA's :52	NA's :1

renewable_ratio

Min. : 1.351

1st Qu.: 5.715

Median : 7.964

Mean :10.329

3rd Qu.:11.408

Max. :34.844

NA's :1

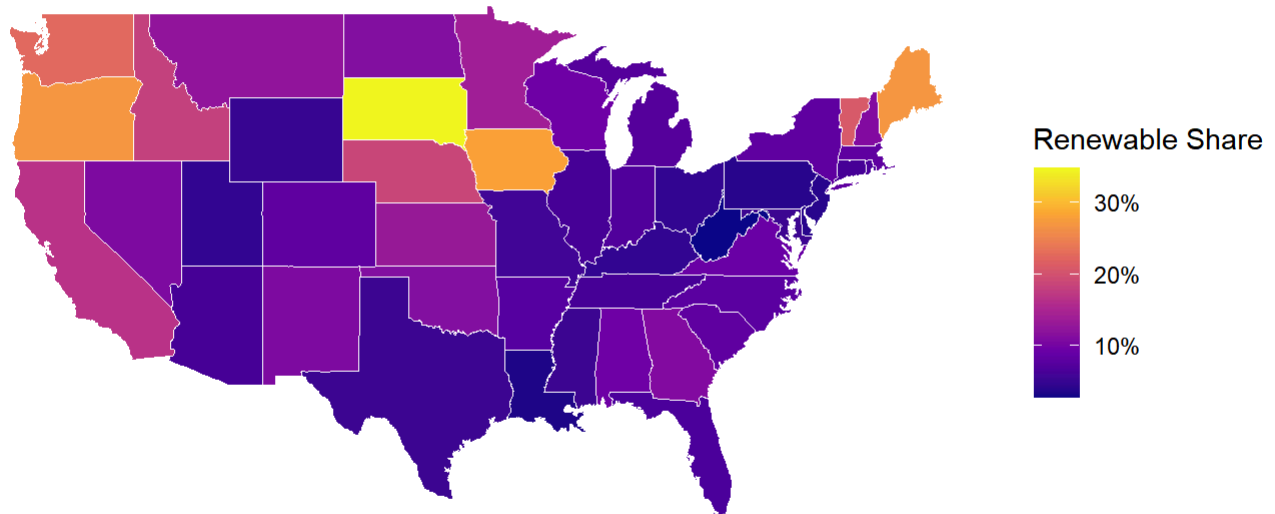
Part 4: Mapping Visualization

```
us_states <- map_data("state")

map_2023 <- us_states |>
left_join(final_2023 |> mutate(region = tolower(state)), by = "region")
ggplot(map_2023, aes(long, lat, group = group, fill = renewable_share)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(option = "C", labels = percent_format(accuracy = 1), name = "Renewable Sh
labs(title = "Renewable Energy Share by State (2023)", subtitle = "DOE data 2021-2023", captio
theme_minimal() +
theme(axis.text = element_blank(), axis.title = element_blank(), panel.grid = element_blank())
```

Renewable Energy Share by State (2023)

DOE data 2021–2023



STAT 133 Project

```
maps_all <- us_states |>
left_join(energy_all |> mutate(region = tolower(state)), by = "region")
```

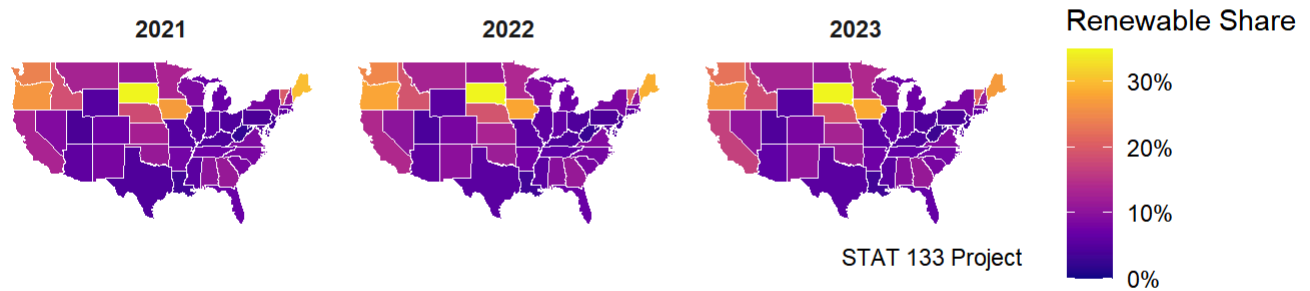
Warning in left_join(us_states, mutate(energy_all, region = tolower(state))), : Detected an unexpected many-to-many relationship between `x` and `y`.

- i Row 1 of `x` matches multiple rows in `y`.
- i Row 1 of `y` matches multiple rows in `x`.
- i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

```
ggplot(maps_all, aes(long, lat, group = group, fill = renewable_share)) +
  geom_polygon(color = "white", linewidth = 0.15) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(option = "C", labels = percent_format(accuracy = 1), limits = c(0, max(ma
  labs(title = "Renewable Energy Share by State (2021–2023)", subtitle = "Faceted by year", capt
  facet_wrap(~year, ncol = 3) +
  theme_minimal() +
  theme(axis.text = element_blank(), axis.title = element_blank(), panel.grid = element_blank(),
```

Renewable Energy Share by State (2021–2023)

Faceted by year



The map shows clear regional differences. States in the Northwest and Midwest, like Washington, Oregon, and Iowa, have the highest renewable energy shares because they use more wind and hydropower. States in the South, like Louisiana and Alabama, still depend on fossil fuels, so their renewable share is much lower.

When we look at EV data, California stands out—it has both a high renewable share and many EVs, showing strong clean energy policies. The map helps show these differences clearly and makes it easy to see how geography and policy affect renewable energy use.